

MODULE 2 | PART 2
NUMBER THEORY
IOQM & RMO 2026

NUMBER THEORY II

Modulo Operation (Modulo of Integers)

The modulo operation finds the remainder when one integer is divided by another.

We write: $a \bmod n = r$
where:

- a is any integer (called the dividend),
- n is the modulus (the number you're dividing by),
- r is the remainder when a is divided by n .

E.g.: $5 \bmod 2 = 1$, because 5 divided by 2 has a quotient of 2 and a remainder of 1.

(Dividend = divisor $\times$ quotient + remainder)

$9 \bmod 3 = 0$, because 9 divided by 3 has a quotient of 3 and a remainder of 0.

Example:

Find $30 \bmod 4$

Solution:

$30 \div 4 = 7$ with remainder 2

$\Rightarrow 30 \bmod 4 = 2$

Example:

Compute $17 \bmod 5$

Solution:

$17 \div 5 = 3$ with remainder 2

$\Rightarrow 17 \bmod 5 = 2$

Example:

Compute $100 \bmod 7$

Solution:

$100 \div 7 = 14$ with remainder 2

$\Rightarrow 100 \bmod 7 = 2$



Example:

Find $-3 \bmod 5$.

Solution:

We want the remainder to be a non-negative number less than 5 (i.e., 0 or a positive number less than 5).

Find the closest number less than or equal to -3 that is divisible by 5.

The multiples of 5 are: $\dots, -10, -5, 0, 5, 10, \dots$

The closest number less than or equal to -3 that is divisible by 5 is -5 .

Now remainder $= -3 - (-5) = 2$

So, $-3 \bmod 5 = 2$

Example:

Find $-7 \bmod 4$

Solution:

The closest number less than or equal to -7 that is divisible by 4 is -8 .

Remainder $= -7 - (-8) = 1$

So, $-7 \bmod 4 = 1$



Congruence of Integers

Two integers a and b are said to be congruent modulo m , denoted by $a \equiv b(\bmod m)$, if a and b have the same remainder when they are divided by a non-zero integer m .

If the remainders are different, then a and b are said to be not congruent modulo m , denoted by $a \not\equiv b(\bmod m)$.

By definition of congruence, $a \equiv b(\bmod m)$ means $a \bmod m = b \bmod m$.

$a \equiv b(\bmod m)$ iff m divides $a - b$.



Basic Properties of Congruence

In what follows n, b, c, d, x, y are integers.

- ❖ $a \equiv a(\bmod n)$
(Reflexive relation for all $a \in \mathbb{Z}$)
- ❖ $a \equiv b(\bmod n) \Leftrightarrow b \equiv a(\bmod n)$
(Symmetric relation for all $a, b \in \mathbb{Z}$)
- ❖ $a \equiv b(\bmod n), b \equiv c(\bmod n)$
 $\Rightarrow a \equiv c(\bmod n)$
(Transitive relation for all integers a, b, c)

❖ If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then

$$(i) a + c \equiv b + d \pmod{n}$$

$$(ii) a - c \equiv b - d \pmod{n}$$

$$(iii) ac \equiv bd \pmod{n}$$

$$(iv) ax + cy \equiv bx + dy \pmod{n}.$$

❖ If $a \equiv b \pmod{n}$, then

$$(i) a + c \equiv b + c \pmod{n}$$

$$(ii) a - c \equiv b - c \pmod{n}$$

$$(iii) ac \equiv bc \pmod{n}$$

$$(iv) a + k_1n \equiv b + k_2n \pmod{n};$$

$$k_1, k_2 \in \mathbb{Z}$$

$$(v) a^m \equiv b^m \pmod{n} \quad m \in \mathbb{N}$$

❖ $a \equiv b \pmod{n}$

$$\Rightarrow f(a) \equiv f(b) \pmod{n}$$

Where f is a polynomial over $\mathbb{Z}$. i.e.,

$f(x)$ is a polynomial with integer coefficients.

❖ $a\lambda \equiv b\lambda \pmod{n}$

$$\Rightarrow a \equiv b \pmod{\frac{n}{\gcd(\lambda, n)}}$$

❖ If $n \neq 0$ and $\gcd(a, n) = 1$, then there exists an integer a' such that $aa' \equiv 1 \pmod{n}$, which is called the inverse of ' a ' modulo n .



Example:

Is $17 \equiv 2 \pmod{5}$?

Solution:

To check whether $17 \equiv 2 \pmod{5}$, we divide 17 by 5.

We get:

$$17 \div 5 = 3 \text{ remainder } 2$$

So, $17 \pmod{5} = 2$. But $2 \pmod{5} = 2$ as well. Since both leave the same remainder when divided by 5, the congruence holds.

Alternatively, subtracting:

$$17 - 2 = 15$$

and since 15 is divisible by 5, this confirms the result.

Thus, $17 \equiv 2 \pmod{5}$

Example:

Find all values of x such that:

$$x \equiv 3 \pmod{5} \text{ and } 0 \leq x < 20$$

Solution:

This means that when x is divided by 5, the remainder is 3.

Or in other words:

$$x = 5k + 3 \text{ for some integer } k$$

We can plug in values of k (starting from 0) and compute $x = 5k + 3$:



$$k: x = 5k + 3$$

$$0: 5(0) + 3 = 3$$

$$1: 5(1) + 3 = 8$$

$$2: 5(2) + 3 = 13$$

$$3: 5(3) + 3 = 18$$

$$x = 3, 8, 13, 18$$

Example:

Find the least positive value of x such that,
 $71 \equiv x \pmod{8}$

Solution:

$$71 \equiv x \pmod{8}$$

$$64 \equiv 0 \pmod{8}$$

$$64 + 7 \equiv 0 + 7 \pmod{8}$$

$$71 \equiv 7 \pmod{8}$$

$$\text{So } x = 7$$

Example:

Solve $5x \equiv 4 \pmod{6}$

Solution:

$$5x - 4 = 6n \text{ for some integer } n$$

$$x = \frac{6n + 4}{5}$$

When we put $n = 1, 6, 11, 16 \dots$ then $6n + 4$ is divisible by 5

$$\text{If } n = 1, \text{ then } x = \frac{6 \times 1 + 4}{5} = 2$$

$$\text{If } n = 6, \text{ then } x = \frac{6 \times 6 + 4}{5} = 8$$

Therefore the solutions are 2, 8, 14, 20...

Example:

$$a \equiv 3 \pmod{7}$$

$$b \equiv 5 \pmod{7}$$

Find: $a + b \pmod{7}$

Solution:

Given,

$$a \equiv 3 \pmod{7} \text{ and } b \equiv 5 \pmod{7}$$

$$a + b \equiv 3 + 5 \pmod{7}$$

$$a + b \equiv 8 \pmod{7}$$

Now reduce 8 modulo 7:

$$8 \div 7 = 1 \text{ remainder } 1 \Rightarrow 8 \pmod{7} = 1$$

Example:

If $x \equiv 1 \pmod{4}$, what is $x^2 \pmod{4}$?

Solution:

We are given:

$$x \equiv 1 \pmod{4}$$

Now, using the multiplication property of congruence, which states:

$$\text{If } a \equiv b \pmod{n}, \text{ then } a \cdot a \equiv b \cdot b \pmod{n}$$

Apply this to square both sides:

$$x^2 \equiv 1^2 \pmod{4}$$

$$x^2 \equiv 1 \pmod{4}$$



Example:

If $a \equiv 4 \pmod{7}$ and $b \equiv 6 \pmod{7}$, what is the remainder when ab is divided by 7?

Solution:

By the multiplication property of congruence, we know:

$$ab \equiv 4 \times 6 \pmod{7} = 24 \pmod{7}$$

Now, divide 24 by 7:

$$24 \div 7 = 3 \text{ remainder } 3$$

$$\text{So, } 24 \pmod{7} = 3$$

The remainder when ab is divided by 7 is 3

Example:

Find the remainder when 2^{81} is divided by 17.

Solution:

$$2^{81} \equiv x \pmod{17}$$

$$\text{we know } 2^3 \equiv 8 \pmod{17}$$

$$(2^3)^3 \equiv 8^3 \pmod{17}$$

$$2^9 \equiv 512 \pmod{17}$$

$$2^9 \equiv 2 \pmod{17}$$

$$(2^9)^9 \equiv 2^9 \pmod{17}$$

$$2^{81} \equiv 512 \pmod{17}$$

$$2^{81} \equiv 2 \pmod{17}$$

So when 2^{81} is divided by 17, then the remainder is 2.

**Example:**

If $x \equiv 3 \pmod{7}$ and $y \equiv 5 \pmod{7}$, find the value of $(3x + 2y) \pmod{7}$

Solution:

$$x \equiv 3 \pmod{7}, y \equiv 5 \pmod{7}$$

$$3x \equiv (3 \times 3) \pmod{7} \equiv 9 \pmod{7}$$

$$\equiv 2 \pmod{7}$$

$$2y \equiv (2 \times 5) \pmod{7} \equiv 10 \pmod{7}$$

$$\equiv 3 \pmod{7}$$

$$\Rightarrow 3x + 2y \equiv (2 + 3) \pmod{7} \equiv 5 \pmod{7}$$

Example:

If $x \equiv 4 \pmod{9}$, find the remainder when $x^2 + 2x + 1$ is divided by 9.

Solution:

This is essentially simplifying a quadratic expression under mod.

Given:

$$x \equiv 4 \pmod{9}$$

Substitute:

$$x^2 + 2x + 1 \equiv (4^2 + 2 \cdot 4 + 1) \pmod{9}$$

$$\equiv 16 + 8 + 1 \pmod{9}$$

$$\equiv 25 \pmod{9}$$

Now reduce:

$$25 \div 9 = 2 \text{ remainder } 7$$

$$\Rightarrow 25 \pmod{9} \equiv 7 \pmod{9}$$

Example:

What is the remainder when 2016^{2016} is divided by 2017?

Solution:

$$\text{As } 2016 \equiv -1 \pmod{2017}$$

$$\Rightarrow 2016^{2016} \equiv 1 \pmod{2017}$$



1. Show that $2^{55} + 1$ is divisible by 11.



2. Find the largest positive integer n such that $n^3 + 100$ is divisible by $(n + 10)$.



3. Find the last two digits of $(56789)^{41}$.



Factorial Notation

❖ Notation: $n!$ (factorial n)

$$n! = n \cdot (n-1) \cdot (n-2) \dots 3 \cdot 2 \cdot 1$$

$$n! = n \cdot (n-1)! \text{ where } n \in \mathbb{N}$$

❖ $0! = 1! = 1$

❖ Factorials of negative integers are not defined.

❖ $n!$ is also denoted by $\lfloor n$

$$❖ n! = n(n-1)! = n(n-1)(n-2)!$$

$$❖ (2n)! = 2^n \cdot n! [1 \cdot 3 \cdot 5 \cdot 7 \dots (2n-1)]$$

$$❖ \frac{n!}{r!} = n(n-1)(n-2) \dots (r+1) \quad [r < n]$$

$$❖ \frac{n!}{(n-r)!} = n(n-1)(n-2) \dots (n-r+1) \quad [r < n]$$

❖ If $x! = y! \Rightarrow x = y$ or

$$x = 0, y = 1 \text{ or } x = 1, y = 0$$

- ❖ Prime factorization of $n!$: Let p be a prime number and n be a positive integer, then exponent of p in the prime factorization of $n!$ is denoted by $E_p(n!)$ and is given by

$$E_p(n!) = \left[\frac{n}{p} \right] + \left[\frac{n}{p^2} \right] + \left[\frac{n}{p^3} \right] + \dots + \left[\frac{n}{p^k} \right]$$

where, $p^k \leq n < p^{k+1}$ and $[x]$ denotes the integral part of x .

If we isolate the power of each prime contained in any number n , then n can be written as $n = 2^{\alpha_1} \cdot 3^{\alpha_2} \cdot 5^{\alpha_3} \cdot 7^{\alpha_4} \dots$, where α_i are whole numbers.

- ❖ The number of trailing zeros in $n!$ is equal to the exponent of 5 in the prime factorization of $n!$.

Example:

What is $5!$?

Solution:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Example:

Simplify $\frac{6!}{4!}$

Solution:

$$\frac{6!}{4!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = 6 \times 5 = 30$$



Example:

Simplify the expression: $\frac{(n+1)! - n!}{n!}$

Solution:

$$\begin{aligned} (n+1)! &= (n+1)n! \\ \Rightarrow \frac{(n+1)n! - n!}{n!} &= \frac{n!(n+1-1)}{n!} \\ &= \frac{n! \cdot n}{n!} = n \end{aligned}$$

Example:

Let $\frac{1}{3!} + \frac{1}{4!} = \frac{\lambda}{5!}$. Find the value of λ .

Solution:

We calculate:

$$\begin{aligned} \frac{1}{3!} &= \frac{1}{6} \\ \frac{1}{4!} &= \frac{1}{24} \\ \frac{1}{6} + \frac{1}{24} &= \frac{4+1}{24} = \frac{5}{24} \end{aligned}$$

Now

$$\frac{5}{24} = \frac{\lambda}{120} \Rightarrow \lambda = \frac{5 \times 120}{24} = 25$$

Example:

If $\frac{n!}{(n-2)!} = 90$, find n .

Solution:

$$\begin{aligned} \frac{n!}{(n-2)!} &= \frac{n(n-1)(n-2)!}{(n-2)!} \\ &= n(n-1) \end{aligned}$$

$$\text{So, } n(n-1) = 90 \Rightarrow n^2 - n - 90 = 0$$

$$\Rightarrow (n-10)(n+9) = 0$$

$$\Rightarrow n = 10 \text{ (positive solution)}$$

Example:

Find the exponent of 3 in 13!

Solution:

The exponent of 3 in 13! (denoted as $E_3(13!)$) is given by:

$$E_3(13!) = \left\lfloor \frac{13}{3} \right\rfloor + \left\lfloor \frac{13}{3^2} \right\rfloor + \left\lfloor \frac{13}{3^3} \right\rfloor + \dots$$

Compute each term:

$$\left\lfloor \frac{13}{3} \right\rfloor = 4, \quad \left\lfloor \frac{13}{9} \right\rfloor = 1, \quad \left\lfloor \frac{13}{27} \right\rfloor = 0$$

So,

$$E_3(13!) = 4 + 1 + 0 = \boxed{5}$$

Example:

How many trailing zeros are there in 100!?

Solution:

Compute each term:

$100 \div 5 = 20$, so the first term is 20.

$100 \div 25 = 4$, so the second term is 4.

$100 \div 125 = 0$, so we stop here.

Now, add them up:

Total exponent of 5 in $100! = 20 + 4 = 24$

$\therefore$ there are 24 trailing zeros in 100!.



4. If $E = \frac{1}{4} \cdot \frac{2}{6} \cdot \frac{3}{8} \cdot \frac{4}{10} \cdots \frac{30}{62} \cdot \frac{31}{64} = 8^x$, then value of x is



5. Find the remainder when $(1!)^2 + (2!)^2 \dots + (100!)^2$ is divided by 1152.



6. Let $n \in \mathbb{N}$ be such that $(n+1)! + (n+2)! = n! \cdot 440$ then number of zeros at the end of $n!$ is equal to



The Units Digit of Powers of Positive Integers

Let P be the units digit of a positive integer a , and n be the positive integer power of a . Then the units digit of a^n is determined by the units digits of P^n , denoted by $U(P^n)$, and the sequence $(U(P^n), n = 1, 2, 3, \dots)$ follows the following rules:

- ❖ The sequence takes constant values for $P = 0, 1, 5, 6$, i.e. $U(P^n)$ does not change as n changes.
- ❖ The sequence is periodic with a period 2 for $P = 4$ or 9.
- ❖ The sequence is periodic with a period 4 for $P = 2, 3, 7, 8$.



Patterns of the Last Digit of a^n

The last digits of a^n have patterns shown in the table below.

For example, when $a = 2$.

$$2^1 = 2 \quad 2^2 = 4 \quad 2^3 = 8 \quad 2^4 = 16 \\ 2^5 = 32 \quad 2^6 = 64 \quad 2^7 = 128 \quad 2^8 = 256, \dots$$

The last digits of 2^n demonstrate a pattern: 2, 4, 8, 6, 2, 4, 8, 6, etc. ...

n	1	2	3	4	Period
1^n	1				1
2^n	2	4	8	6	4
3^n	3	9	7	1	4
4^n	4	6			2
5^n	5				1
6^n	6				1
7^n	7	9	3	1	4
8^n	8	4	2	6	4
9^n	9	1			2

For example:

- (i) In $(32)^{33}$ unit digit is equal to the unit digit of 2^1 , i.e., 2.
- (ii) In $(23)^{15}$ unit digit is equal to the unit digit of 3^3 , i.e., 7.
- (iii) In $(57)^9$ unit digit is equal to the unit digit of 7^1 , i.e., 7.
- (iv) In $(678)^{22}$ unit digit is equal to the unit digit of 8^2 , i.e., 4.



Example:

What is the unit digit in 7^{22} ?

Solution:

The pattern for the last digit is:

7, 9, 3, 1, 7, 9, 3, 1, etc.

When 22 is divided by 4, the quotient is 5 and the remainder is 2. The last digit of 7^{22} is the same the last digit of 7^2 , so the last digit is 9.

Example:

Find the last digit of

(i) 3^{57}

(ii) 13^{59}

Solution:

(i) The cyclicity of 3 is 4 and $\frac{57}{4}$ gives the remainder 1. So, the last digit of 3^{57} is same as the last digit of 3^1 , i.e., 3.

(ii) The number of digits in the base will not make a difference to the last digit. It is last digit of the base that decides the last digit of the number. For 13^{59} , we find that $\frac{59}{4}$ gives a remainder 3. So the last digit of 13^{59} is same as the last digit of 3^3 , i.e., 7.

**Example:**

What will be the last digit of $(73)^{75^{64^{76}}}$

Solution:

$$\text{Let } (73)^{75^{64^{76}}} = (73)^x$$

$$\text{where } x = 75^{64^{76}} = (75)^{\text{even power}}$$

$\therefore$ Cyclicity of 3 is 4, to find the last digit, we have to find the remainder when x is divided by 4.

$$x = (75)^{\text{even power}} = (76 - 1)^{\text{even power}}$$

So when x is divided by 4 the remainder will be 1.

Therefore, the last digit of $(73)^{75^{64^{76}}}$ will be $3^1 = 3$.

Example:

Find the units digit of

$$x = (35)^{87} + (93)^{46} + (44)^{91}.$$

Solution:

$$x = 35^{87} + 93^{46} + 44^{91}$$

$$= 35^{87} + 93^{(44+2)} + 44^{(90+1)}$$

Cyclicity of 3 is 4, 5 is 1 and 4 is 2.

$\therefore$ The units digit of x is 8



7. Find the units digit of

$$(2^2 + 4^2 + 6^2 + \dots + 100^2)^{64}.$$



8. Find the units digit of the number equal to sum of $1! + 2! + 3! + 4! + \dots + 100!$



9. Find the units digit of the number equal to sum of $3! + (3!)^2 + (3!)^3 \dots + (3!)^{100}$.



Fermat's Little Theorem

Fermat's Little Theorem states that if p is a prime number, then for any integer a , the number $a^p - a$ is an integer multiple of p . In the notation of modular arithmetic, this is expressed as $a^p \equiv a \pmod{p}$.

For example, if $a = 2$ and $p = 7$, then $2^7 = 128$, and $128 - 2 = 126 = 7 \times 18$ is an integer multiple of 7.

If a is not divisible by p , Fermat's Little Theorem is equivalent to the statement that $a^{p-1} - 1$ is an integer multiple of p , or in symbols: $a^{p-1} \equiv 1 \pmod{p}$

For example, if $a = 2$ and $p = 7$, then $2^6 = 64$, and $64 - 1 = 63 = 7 \times 9$ is a multiple of 7.

Example:

Find the remainder when 2^{10} is divided by 11.

Solution:

11 is a prime number.

Since 2 is not divisible by 11, we can apply Fermat's Little Theorem.

$$2^{10} \equiv 1 \pmod{11}$$



Example:

Compute $3^{100} \bmod 7$

Solution:

7 is prime, and 3 is not divisible by 7.

By Fermat's Little Theorem:

$$3^6 \equiv 1 \pmod{7}$$

So:

$$3^{100} = 3^{(6 \cdot 16 + 4)} = (3^6)^{16} \cdot 3^4$$

$$\equiv (1^{16} \cdot 3^4) \pmod{7}$$

$$\equiv 3^4 \pmod{7}$$

Compute $3^4 = 81$

$$81 \bmod 7 = \boxed{4}$$

Example:

Evaluate $5^{123} \bmod 13$

Solution:

13 is prime, and 5 is not divisible by 13.

By Fermat's Little Theorem:

$$5^{12} \equiv 1 \pmod{13}$$

Now:

$$123 = 10 \cdot 12 + 3$$

$$\Rightarrow 5^{123} = (5^{12})^{10} \cdot 5^3 \equiv 1^{10} \cdot 5^3 \pmod{13}$$

$$\equiv 5^3 \pmod{13}$$

$$5^3 = 125, \quad 125 \bmod 13 = 8$$



Euler's Totient Function

❖ The number of positive integers less than or equal to n that are co-prime to n is denoted by $\phi(n)$ and is called Euler's totient function.

❖ Euler's totient function is multiplicative, i.e., if $\gcd(m, n) = 1$, then $\phi(mn) = \phi(m) \cdot \phi(n)$.

❖ For prime p ,

$$\phi(p) = p - 1,$$

$$\begin{aligned} \phi(p^k) &= p^k - p^{k-1} \\ &= p^{k-1}(p - 1). \end{aligned}$$

❖ $\phi(1) = 1$.

$$\phi(n) = n \cdot \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

i.e., If $n = p_1^{a_1} \cdot p_2^{a_2} \cdot \dots \cdot p_k^{a_k}$ then

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$$

Example:

Find $\phi(7)$

Solution:

7 is prime.

$$\text{So, } \phi(7) = 7 - 1 = 6$$



Example:

Find the value of $\phi(9)$.

Solution:

Numbers less than 9 : 1, 2, 3, 4, 5, 6, 7, 8

Coprime with 9 : 1, 2, 4, 5, 7, 8 $\rightarrow$ 6 numbers

$$\phi(9) = 6$$

This can also be obtained by formula.

$$\text{If } n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

we have,

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right)$$

Here, $n = 9$

$$\text{and } 9 = 3^2 \Rightarrow p_1 = 3$$

$$\phi(9) = 9 \left(1 - \frac{1}{3}\right) = 9 \cdot \frac{2}{3} = 6$$

Example:

Find $\phi(20)$ using the formula.

Solution:

Prime factorization of 20:

$$20 = 2^2 \times 5$$

Use Euler's formula:

$$\begin{aligned} \phi(20) &= 20 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) \\ &= 20 \times \frac{1}{2} \times \frac{4}{5} = 20 \times \frac{4}{10} = 8 \end{aligned}$$

**Example:**

Find $\phi(35)$

Solution:

Here $n = 35$

$$n = 7 \times 5 \quad (\text{product of two primes})$$

$$\text{So, } \phi(n) = \phi(7)\phi(5)$$

$$\phi(35) = (7 - 1)(5 - 1)$$

$$= 6 \times 4$$

$$\phi(35) = 24$$

Example:

Find $\phi(1000)$.

Solution:

$$\text{Here } n = 1000 = 2^3 \times 5^3.$$

Distinct prime factors are 2 and 5.

$$\phi(n) = n \times \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots$$

$$\phi(1000) = 1000 \times \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right)$$

$$\phi(1000) = 1000 \times \left(\frac{1}{2}\right) \left(\frac{4}{5}\right)$$

$$\phi(1000) = 400$$

Euler's Theorem

$$\text{If } \gcd(a, n) = 1, \text{ then } a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$ is Euler's totient function, giving the count of numbers less than n that are relatively prime to n .

Example:

Show that Euler's Theorem holds true for $a = 3$ and $n = 10$.

Solution:

Given, $a = 3, n = 10 \Rightarrow \gcd(3, 10) = 1$

Then $a^{\phi(n)} \equiv 1 \pmod{n}$

$3^{\phi(10)} \equiv 1 \pmod{10}$,

$\phi(10) = 4$

$\Rightarrow 3^4 \equiv 1 \pmod{10}$

$\Rightarrow 81 \equiv 1 \pmod{10}$

$\therefore$ Euler's Theorem holds good

Example:

Using Euler's Theorem, simplify $6^{100} \pmod{35}$.

Solution:

Check $\gcd(6, 35) = 1$, so we proceed.

Since $35 = 5 \times 7$, we calculate:

$$\begin{aligned}\phi(35) &= 35 \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{7}\right) \\ &= 35 \cdot \frac{4}{5} \cdot \frac{6}{7} = 24\end{aligned}$$

So: $6^{24} \equiv 1 \pmod{35}$

$\Rightarrow 6^{100} = (6^{24})^4 \cdot 6^4 \equiv 1^4 \cdot 6^4 \pmod{35}$

Now compute $6^4 = 1296$, and

$1296 \pmod{35} = 1296 - 35 \cdot 37 = 1$

**Example:**

What is the remainder when 4^{117} is divided by 25?

Solution:

$\gcd(4, 25) = 1 \Rightarrow$ Euler's Theorem holds.

$$\phi(25) = 25 \left(1 - \frac{1}{5}\right) = 25 \cdot \frac{4}{5} = 20$$

By Euler's Theorem, $4^{\phi(25)} \equiv 1 \pmod{25}$

$\Rightarrow 4^{20} \equiv 1 \pmod{25}$

Now, $117 \pmod{20} = 17$

$\Rightarrow 4^{117} \equiv 4^{17} \pmod{25}$

Also, $4^4 = 256$ and $256 \pmod{25} = 6$

i.e. $4^4 \pmod{25} = 6 \Rightarrow 4^4 \equiv 6 \pmod{25}$

$\Rightarrow 4^8 \equiv 6^2 \pmod{25}$

$\Rightarrow 4^8 \equiv 11 \pmod{25}$

$\Rightarrow 4^{16} \equiv 121 \pmod{25}$

$\Rightarrow 4^{16} \equiv 21 \pmod{25}$

$\Rightarrow 4^{17} \equiv 21 \cdot 4 \pmod{25}$

$\Rightarrow 4^{17} \equiv 84 \pmod{25}$

$\Rightarrow 4^{17} \equiv 9 \pmod{25}$

So, $4^{117} \equiv 9 \pmod{25}$

Remainder = 9

Wilson's Theorem

Natural number $n \geq 2$ is a prime number if and only if $(n-1)! \equiv -1 \pmod{n}$.

Which also implies,

$$(n-1)! \pmod{n} = (n-1)$$

Example:

What will be the remainder when $568!$ is divided by 569 ?

According to Wilson's theorem, we have,
for prime number ' p ',

$$\Rightarrow (p-1)! \bmod p = (p-1)$$

In this case, 569 is a prime number.

Thus,

$$\Rightarrow 568! \bmod 569 = 568.$$

Hence, when $568!$ is divided by 569 , we get 568 as the remainder.

Example:

Prove $(p-2)! \equiv 1 \pmod{p}$, when p is prime.

Solution:

According to Wilson's theorem, we have,
for prime number ' p ',

$$\Rightarrow (p-1)! \bmod p = (p-1)$$

Now observe:

$$(p-1)! = (p-1)(p-2)!$$

So, substituting into Wilson's Theorem:

$$(p-1)(p-2)! \equiv -1 \pmod{p}$$

Note that $p-1 \equiv -1 \pmod{p}$.

So:

$$(-1)(p-2)! \equiv -1 \pmod{p}$$

Multiply both sides by -1 :

$$(p-2)! \equiv 1 \pmod{p}$$

Example:

What is the remainder when $10!$ is divided by 13 ?

Solution:

By Wilson's Theorem

$$12! \equiv -1 \pmod{13}$$

$$12! \equiv (-1 + 13) \pmod{13}$$

$$11! \equiv 1 \pmod{13} \text{ by 'dividing' by } 12$$

$$11! \equiv 1 + 5 \times 13 \pmod{13}$$

$$10! \equiv 6 \pmod{13} \text{ by 'dividing' by } 11$$

Example:

Find the remainder when $2016! - 2015!$ is divided by 2017 .

Solution:

$$2015! \equiv 2016! \cdot 2016^{-1} \pmod{2017}$$

$$\equiv (-1) \cdot (-1)^{-1} \pmod{2017}$$

$$\equiv (-1) \cdot (-1) \pmod{2017}$$

$$\equiv 1 \pmod{2017}$$

$$\text{Thus, } 2016! - 2015! \equiv -1 - 1 \pmod{2017}$$

$$\equiv 2015 \pmod{2017}.$$



Base System

The number system that we work in is called the 'decimal system'. This is because there are 10 digits in the system 0 – 9. There can be alternative system that can be used for arithmetic operations. Some of the most commonly used systems are: binary, octal, and hexadecimal.

These systems find applications in computing.

Binary system [2] has 2 digits: 0, 1.

Octal system [8] has 8 digits: 0, 1, ..., 7.

Hexadecimal system [16] has 16 digits: 0, 1, 2, ..., 9, A, B, C, D, E, F

After 9, we use the letters to indicate digits. For instance, A has a value 10, B has a value 11, C has a value 12, and so on. This principle applies to all base systems.

The counting sequences in each of the systems would be different though they follow the same principle.

Conversion

Conversion of numbers from

- (i) decimal system to other base system.
- (ii) other base system to decimal system.



Conversion from Base 10 to any Other Base

Example:

Convert $(122)_{10}$ to base 8 system.

Solution:

The number in decimal is consecutively divided by the number of the base to which we are converting the decimal number. Then list down all the remainders in the reverse sequence to get the number in that base.

8	122	
8	15	2
8	1	7
	0	1

So, here $(122)_{10} = (172)_8$.

Example:

Convert $(169)_{10}$ in base 7.

Solution:

7	169	
7	24	1
7	3	3
	0	3

$(169)_{10} = (331)_7$



Example:

Convert $(0.3125)_{10}$ to binary equivalent.

Solution:

$$0.3125 \times 2 = 0.625 \Rightarrow 0$$

$$0.625 \times 2 = 1.25 \Rightarrow 1$$

$$0.25 \times 2 = 0.5 \Rightarrow 0$$

$$0.5 \times 2 = 1.0 = 1$$

So, $(0.3125)_{10} = (0.0101)_2$

Example:

Convert $(1987.725)_{10} \rightarrow (\dots\dots\dots)_8$

Solution:

First convert non-decimal part into base 8.

8	1987	
8	248	3
8	31	0
8	3	7
	0	3

$$\therefore (1987)_{10} = (3703)_8$$

Now we have to convert $(0.725)_{10}$ to base 8

Multiply the fractional part by 8.

Keep on accomplishing integral parts after multiplication with decimal part till decimal part is zero, or the required precision is reached

$0.725 \times 8 = [5.8]$	5
$0.8 \times 8 = [6.4]$	6
$0.4 \times 8 = [3.2]$	3
$0.2 \times 8 = [1.6]$	1
$0.6 \times 8 = [4.8]$	4

$$\therefore (0.725)_{10} = (0.56314\dots)_8$$

$$\therefore (1987.725)_{10} = (3703.56314\dots)_8$$

Conversion from any Other Base to Decimal System

Example:

Convert $(231)_8$ into decimal system.

Solution:

To convert $(231)_8$ to base 10, we determine the value of each digit based on its position as follows:

$$1 \rightarrow 8^0 \times 1$$

$$3 \rightarrow 8^1 \times 3$$

$$2 \rightarrow 8^2 \times 2$$

Hence,

$$(231)_8 = (8^0 \times 1 + 8^1 \times 3 + 8^2 \times 2)_{10}$$

$$(231)_8 = (1 + 24 + 128)_{10}$$

$$(231)_8 = (153)_{10}$$

Example:

Convert $(761.56)_8 \rightarrow (\dots\dots\dots)_{16}$

Solution:

In such conversion, which are standard form conversions, it is easier to

$$(761.56)_8 \rightarrow (\dots\dots\dots)_2 \rightarrow (\dots\dots\dots)_{16}$$

Converting every digit in base 8 to base 2,

$$(111110001.101110)_2 \rightarrow (1F1.B8)_{16}$$



Example:

Convert $(3C8.08)_{16}$ to decimal

Solution:

$$(3C8.08)_{16} = 3 \times 16^2 + C \times 16^1 + 8 \times 16^0 + 0 \times 16^{-1} + 8 \times 16^{-2}$$

$$= 768 + 192 + 8 + 0 + 0.03125$$

$$= (968.03125)_{10}$$

$$\text{So, } (3C8.08)_{16} = (968.03125)_{10}$$



10. A number N in base 10 is 503 in base b and 305 in base $b + 2$. What is the product of the digits of N ?

[IOQM 2021]



Digit Sum Characteristic Theorem

The sum of digits of a number is congruent to the number modulo 9. The same holds for modulo 3.

Proof: Since $10^n \equiv 1 \pmod{9}$ for all $n \in \mathbb{N}$, any number written in decimal representation such as $(a_n a_{n-1} a_{n-2} \cdots a_1 a_0)_{10}$ is congruent to $(a_n + a_{n-1} + \cdots + a_1 + a_0) \pmod{9}$

Example:

Let $N = 5 \times 10^5 + 4 \times 10^4 + 3 \times 10^3 + 2 \times 10^2 + 1 \times 10 + 7$. Find $N \pmod{9}$

Solution:

$$N = 5 \times 10^5 + 4 \times 10^4 + 3 \times 10^3 + 2 \times 10^2 + 1 \times 10 + 7$$

$$= 543217$$

$$\begin{aligned} \text{Sum of digits of } N &= 5 + 4 + 3 + 2 + 1 + 7 \\ &= 22 \end{aligned}$$

By digit sum characteristic theorem,

$$N \equiv 22 \pmod{9}$$

$$N \equiv 4 \pmod{9}$$

$$\Rightarrow N \pmod{9} = 4$$

Example:

Let a 6-digit number $N = 4abc32$ where a, b, c are digits. If $N \equiv 0 \pmod{9}$, find the possible values of $a + b + c$.

Solution:

$$\begin{aligned} N &\equiv 4 + a + b + c + 3 + 2 = a + b + c + 9 \\ &\equiv 0 \pmod{9} \end{aligned}$$

$$\Rightarrow a + b + c + 9 \equiv 0 \pmod{9}$$

$$\Rightarrow a + b + c \equiv 0 \pmod{9}$$

So $a + b + c$ must be a multiple of 9.

But since a, b, c are digits their values can range from 0 to 9

$$\text{max total} = 9 + 9 + 9 = 27$$

So possible values of $a + b + c$: 0, 9, 18, or 27

Example:

A number N is divisible by 9. If one digit is removed and the new number has digit sum 25, find the possible value(s) of the digit that was removed.

Solution:

Since $N \equiv 0 \pmod{9}$, the total digit sum of N must be a multiple of 9.

Let the removed digit be d , so

$$25 + d \equiv 0 \pmod{9}$$

$$\Rightarrow d \equiv -25 \pmod{9}$$

$$\Rightarrow d \equiv 2 \pmod{9}$$

Smallest positive such $d \in [0, 9]$ is 2

Example:

Find the remainder when 4333^{3333} is divided by 9.

Solution:

$$4333 \equiv 4 + 3 + 3 + 3 \pmod{9}$$

$$\equiv 13 \pmod{9}$$

$$\equiv 4 \pmod{9}$$

$$\therefore 4333^3 \equiv 4^3 \pmod{9}$$

$$\equiv 64 \pmod{9}$$

$$\equiv 1 \pmod{9}$$

$$\equiv 4333^{3333} \equiv 1 \pmod{9}$$

i.e., when 4333^{3333} is divided by 9, the remainder is 1.



11. Let $S(M)$ denote the sum of the digits of a positive integer M written in base 10. Let N be the smallest positive integer such that $S(N) = 2013$. What is the value of $S(5N + 2013)$?

[PRMO 2013]



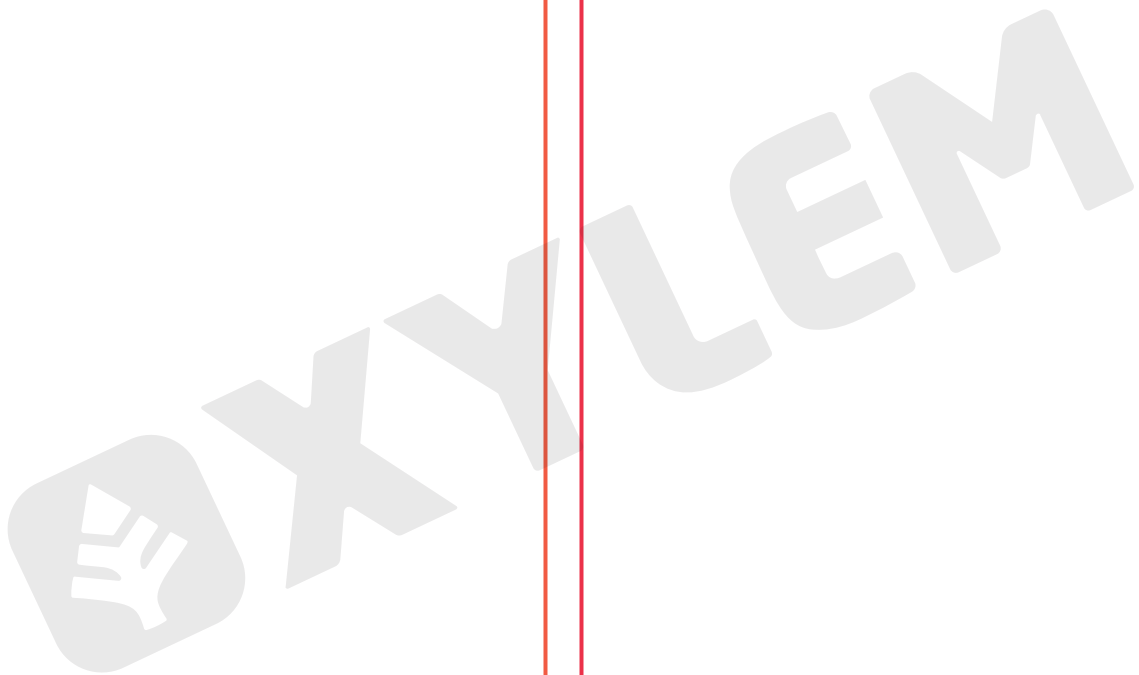
ANSWER KEY

(1)	(4) -12	(7) 0	(10) 64
(2) 890	(5) 41	(8) 3	(11) 18
(3) 89	(6) 3	(9) 0	

Space for rough work



Space for rough work



ALL COURSES IN THE EDUCATION SECTOR ALL IN ONE PLATFORM

XYLEM



NEET/JEE REPEATERS PROGRAMME

☎ 6009 100 300



TWO-YEAR
PREMIER ENTRANCE
COACHING PROGRAMME
☎ 6009 100 300

AIIMS JIPMER
AFMC IIT NIT
IISER IISc IIST



☎ 9995 400 800

OFFLINE FOUNDATION
CLASSES FOR 8-10
STUDENTS IN THE
CBSE CURRICULUM

ONLINE FOUNDATION
CLASSES FOR
6-10 STUDENTS
(STATE, CBSE AND ICSE)

TUITION FOR 9-12

OFFLINE TUITION CENTERS ☎ 6009 100 300

ONLINE TUITION

FOR 4-12 ☎ 9995 400 800

CLASS 11+ NEET - JEE

CLASS 12+ NEET - JEE

ONLINE COACHING STATE & CBSE

☎ 6009 100 300

PSC OFFLINE & ONLINE YOUR SUCCESS PARTNER

☎ 9995 100 800

IAS OFFLINE RESIDENTIAL & ONLINE BATCHES

FROM ASPIRATION TO ADMINISTRATION

☎ 8714 418 777

XYLEM COMMERCE PRO

KOZHIKODE

OFFLINE & ONLINE

☎ 8129 800 100



CHARTERED
ACCOUNTANCY



ASSOCIATION OF CHARTERED
CERTIFIED ACCOUNTANTS

B.Com + **ACCA**



COST AND MANAGEMENT
ACCOUNTANT



CERTIFIED MANAGEMENT
ACCOUNTANT

CENTRAL GOVERNMENT EXAMS

☎ 8714 418 777

- STAFF SELECTION COMMISSION (SSC)
- RAILWAY RECRUITMENT BOARD (RRB)
- BANKING SECTOR & INSURANCE SECTOR

XYLEMONLINE STORE

BOOKS AVAILABLE AT THE XYLEM STORE

NEET, JEE, KEAM BOOKS
GOVT. EXAM BOOKS, PSC RANK FILES & RRB

CA, ACCA, CMA BOOKS, CUET BOOKS
SCHOOL PREPARATION BOOKS



www.xylem.live/store

